

ASHI SINHA she/her

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Education

University of Massachusetts Amherst, US

Aug 2024 – May 2026

Master of Science in Computer Science (GPA: 3.97/4.0)

Coursework: Advanced Algorithms, Distributed Systems, Systems for Data Science, Machine Learning, Advanced NLP, Reinforcement Learning

Birla Institute of Technology and Science, Pilani, India

Aug 2017 – May 2022

Bachelor of Engineering in Computer Science + Master of Science in Physics (GPA: 8.2/10)

Coursework: Operating Systems, Object Oriented Programming, Data Structures and Algorithms, Database Management, Software Engineering

Work Experience

IBM Research

Jan 2026 – Present

Graduate Researcher

- Recast deep research over data lakes as a budgeted search problem and built **Baikal**, using **embedding-based clustering** to organize **13K+** tables and **240K** passages into semantic regions.
- Formulated region selection as a **multi-armed bandit** balancing **exploration-exploitation**, using finding quality as a reward to update value estimates under ϵ -greedy and **UCB** policies.
- Designed an **LLM-as-judge** rubric (groundedness, relevance, distinctness, utility) scored by GPT-5-mini on HybridQA and TAT-QA, beating DeepSearcher and OpenCode baselines by **28%** and **36%**. Submitted to **AAAI 2027**.
- Conducted through the CS 698DS Industry Practicum at UMass Amherst, under the guidance of Prof. Andrew McCallum.

Amazon

Jul 2022 – Jul 2023

Software Development Engineer

- Implemented In-App Purchase (IAP) for Amazon Appstore using **React Native** and **multi-threaded C++**, reducing purchase-flow latency by **40%**.
- Integrated deep-link routing into the purchase workflow, enabling users to enter checkout directly from app and product surfaces and reducing time-to-purchase by **55%** during peak periods.
- Expanded **GTest** unit-testing coverage across core IAP transaction paths and edge cases, increasing code coverage by **70%** and reducing regression risk in production releases.

Software Development Intern

Jan 2022 – Jun 2022

- Developed a **Java/Android** service from scratch that ranks installed FireTV apps by a multi-factor model (popularity, last-used time, storage size) and surfaces the top 5 deletion candidates, freeing an average of **5 GB** per device.
- Deployed the service on **AWS ECS/Fargate**, integrating with internal build and version-management tooling to support scalable production releases.
- Resolved **20 bugs** in the FireTV Android codebase through 12 code reviews and 8 server-side changes, improving app stability and shipping fixes to production.

Microsoft

May 2021 – Jul 2021

Software Development Intern

- Engineered an **NLP** clustering pipeline in **C#** and **Python** for the Bing search team's *Triggering Expansion with Domain Graphs* initiative, grouping **1,000+** semantically similar queries and improving relevance for ambiguous searches by **25%** based on NDCG evaluation on held-out queries.

Projects

Clinical Reporting Platform | *One Vector (Early-stage Health-Tech Startup)* | **React, Flask, AWS**

Jun 2025 – Aug 2025

- Automated Development Safety Update Report (DSUR) generation, reducing manual reporting effort by **85%** via multi-section **React/TypeScript** forms with versioning, backed by **Pandoc, DynamoDB**, and **S3**.
- Delivered 8+ platform features, including secure **Flask** document-access **APIs** on **AWS** using **presigned S3 URLs** and caching, reducing file retrieval latency by **35%**.
- Integrated a **RAG**-based medical document search workflow with **OpenSearch** and **GPT-4**, allowing users to search, retrieve, and summarize relevant medical documents within the platform.

RepoRadar – Codebase Summarizer (**HackUMass XIII**) | **Python, AST, AWS, LLM APIs, Pyvis, NetworkX**

- Built an AI-powered codebase summarization tool that helps developers understand large Python repositories by extracting imports, functions, classes, and file-level dependencies from repos containing **150+** files.
- Engineered an end-to-end pipeline that parses Python repos with **AST**, chunks code, and generates schema-constrained **LLM** summaries with **interactive Pyvis/NetworkX** dependency graphs.
- Deployed the tool on **AWS Elastic Beanstalk** with caching, reducing request latency by **80%** and API costs by **40%**.

Recommender Companion | *Full-Stack Development, Angular, Flask, GPT, Docker, MySQL*

- Built a full-stack book and movie recommendation web app with an **Angular** frontend and **Flask** REST API, generating personalized GPT-based recommendations from user preferences and interaction history.
- Integrated Microsoft **OAuth**, **MySQL**-backed user/profile storage, Dockerized services, and **Flyway** migrations, enabling reproducible environment setup in under 5 minutes.

Technical Skills

Languages & Frameworks: C++, C#, Java, Python, JavaScript, SQL, React, Angular, Node.js, Flask, Spring, .NET

Data & Cloud: AWS, Docker, Kubernetes, MySQL, DynamoDB, OpenSearch, Spark, Hadoop, Kafka, MapReduce

AI/ML: PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, LLMs, Generative AI, RAG, NLP

Dev Tools: Git, JIRA, VS Code, Android Studio, Eclipse, JUnit, pytest, Google Test, Cursor, Copilot, Codex